

The Editors
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Dear Editors,

We submit the manuscript, “Reliable comparison of severe felt-intensity classifiers with versioned and dependent event data: a USGS DYFI benchmark,” for consideration as a Research Article.

Engineering model selection can be invalidated by catalogue revision, outcome-derived inputs, dependent events crossing data splits, temporal contamination, or unresolved point-score differences. The manuscript asks whether performance and model-superiority claims remain supportable when these risks are controlled. It contributes a released retrospective qualification workflow: declared candidates first pass schema, proxy, role, and temporal audits, then are scored on a fixed holdout and compared with paired sequence-cluster uncertainty.

The frozen $M \geq 5$ snapshot contains 7,520 source records, 2,362 eligible events, and a 430-event temporal holdout. The worked qualification excludes one candidate outside the declared source-only schema, supports all four admissible learned baselines over no-skill, and shows that adding depth improves on a magnitude-only logistic model. The source-only logistic model and XGBoost both round to Brier 0.181, and their paired comparison remains unresolved. The simpler logistic model therefore provides a strong, transparent reference for future methods without being declared superior. These results show which candidates are valid to compare, whether added complexity has produced a measurable improvement, and what performance bar future severe-intensity models should meet.

The article fits the journal’s explicit scope in artificial and machine intelligence, computer science, computational methods, modelling and simulation, seismic evaluation, and engineering research methods. It follows the decision-oriented form of benchmark and engineering machine-learning comparison articles already published by the journal.

Data, code, frozen results, and reproduction instructions are available at <https://github.com/aksha74/dyfi-benchmark-reliability>. A clean clone passes the released validators and benchmark test suite and reconstructs the reported results. The scientific core was frozen in release v1.0.27; JEAS release v1.2.4 packages that unchanged core with engineering-use assets derived from the same frozen inputs.

The manuscript is original and is not under consideration elsewhere. Both authors approved the manuscript and this submission. They declare no competing interests and no specific funding. The study uses aggregate public-domain event metadata, involves no human participants, and redistributes no respondent-level identifying information. Generative-AI assistance is disclosed and all assisted work was reviewed and verified by the authors. The public repository contains a non-peer-reviewed manuscript copy and source, disclosed as a preprint without a DOI; the software and derived data retain their MIT and CC0 terms.

Sincerely,

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